

Machine Learning Based Failure Prediction for Optical Network Digital Twins

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Abstract: We leverage binary classification models to provide digital twins with forecasting ability, to proactively mitigate the impact of failures on availability. Our preferred model achieves 98.5% accuracy on data collected from a live production network. © 2024 The Author(s)

1. Introduction

With ever-increasing dependence on high-speed communications comes higher constraints on optical network availability, calling for cost-effective improvements. This is partly achieved with modern health analytics tools [1], which help reducing the mean time to repair (MTTR) [2]. The next step is to predict failures, to further mitigate their impact on service through appropriate actions. Allowing time-sensitive emulation of the physical network's current state, a digital twin can ideally leverage live monitoring data to predict future states [3]. Since most failures are due to random events impacting fiber cables [4], statistical inference is essential, and machine learning (ML) appears as an effective tool to combine inference with early signs detection [5], thus streamlining the design of failure prediction tools. In this paper, we propose a ML-based model to predict failures within a digital twin. We use field trial data from a live production network to compare and discuss the suitability of prediction models.

2. Field Trial Topology and Digital Twin Architecture

The field trial topology is represented in Fig. 1(a). We monitored a total of 4 lightpaths in both directions (AZ: A to Z and ZA: Z to A). Each optical channel carries net 100 Gbit/s in PM-QPSK modulation format. The 7 fiber segments connecting routing nodes are mixed aerial/buried fiber cables. We collected data over 11 months with a one-second-period. For each channel, the collected data is: pre-FEC BER, post-FEC BER, input/output optical power, chromatic dispersion (CD), and differential group delay (DGD). We further collected cards temperatures. We consider the digital twin architecture described in Fig. 1(b). The twinned network is represented as a collection of transponders, amplifiers, fiber spans, nodes components, etc. Its twin contains a database to record optical performance monitoring (OPM) values and associated timestamps. In our trial, we solely collected OPM data from the receivers. The QoT estimation block combines static inputs, calibration data and parameter estimation based on OPM values to provide GSNR estimations for all active lightpaths. It self-trains to emulate the measured GSNR [6], to detect anomalies when measurements diverge estimations. This information is processed in the health analytics block along with all OPMs to inform the control plane on the current health status. Some health indicators along with selected OPMs are also provided to the failure prediction block to infer future health status.

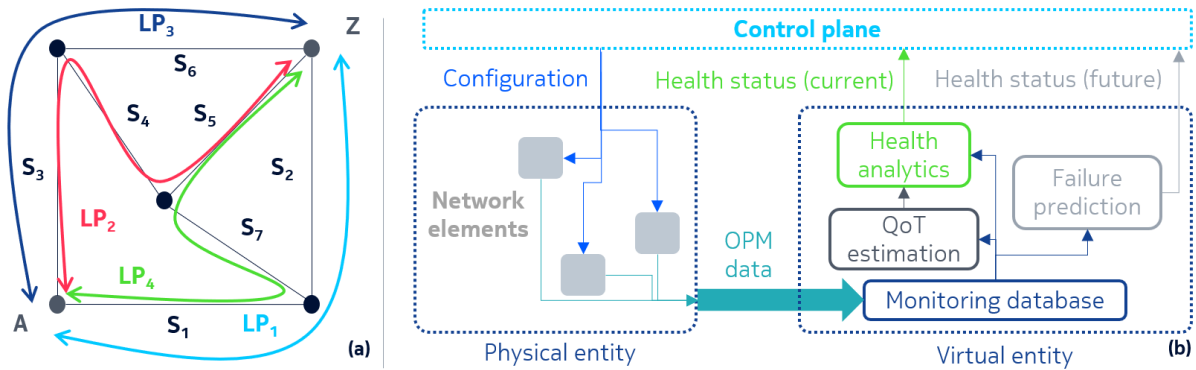


Fig. 1. (a) Field trial topology (b) Digital twin architecture for failure prediction

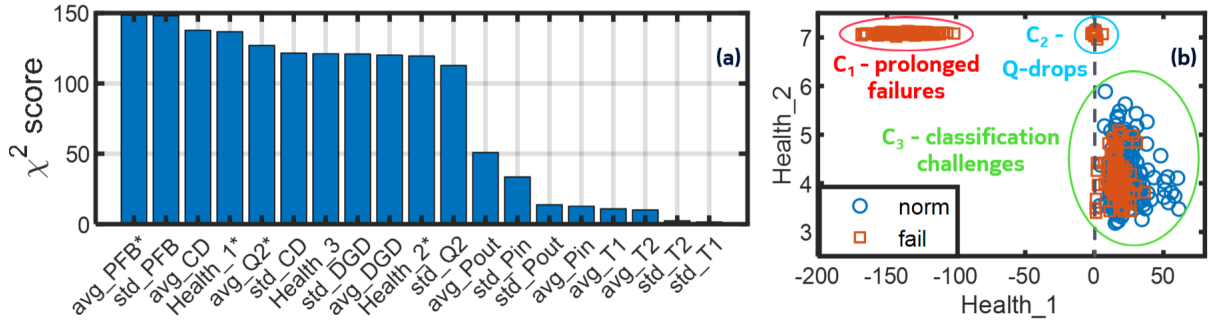


Fig. 2. (a) Features ranked according to χ^2 test, (b) Class separation according to health indicators.

3. Methodology and Feature Selection

We aim to predict - for the channel with the highest failure count - whether or not there will be packet losses in the next hour based on the OPM data from the previous hour. This converts failure prediction into a binary classification problem. Thus, we split the timeline into 7325 one-hour-windows. A window is labeled “fail” if it contains at least one non-zero post-FEC BER value, and “norm” otherwise. Then, classifiers must predict for each window the correct label based on features calculated over the previous window. Most features introduced are averages and standard deviations of the collected OPMs. We simply converted the pre-FEC-BER into a Q^2 margin value in dB, accounting for the FEC limit. We also included two handcrafted features, defined as:

$$\text{Health}_1 = \frac{\text{mean}(Q^2 - Q_{FEC}^2)}{\text{std}(Q^2)} \quad \parallel \quad \text{Health}_2 = \max\left(\left|\frac{Q^2 - \text{movmed}(Q^2)}{\text{movstd}(Q^2)}\right|\right)$$

Health_1 describes the risk of failures due to a combination of QoT fluctuations [7] and relatively low margin, higher risk corresponding to lower, potentially negative, values. *Health_2* captures how far-off is the major Q^2 outlier observed in the one-hour-window. Indeed, Q^2 outliers, also called Q-drops, have been noted to be strongly correlated with failures [8]. Finally, *Health_3* is introduced as the percentage of off-time. The role of these handcrafted features is to simplify the classifiers’ task by providing values which would otherwise require to process a much greater amount of data and/or to perform nonlinear operations for an equivalent result.

We build two datasets to address the large class imbalance of 1:24 “fail” to “norm”. In the training dataset, we include all “fail” samples and complete with an equal amount of randomly selected “norm” samples. The test dataset is the full dataset and is used to properly assess the classifiers performance in realistic conditions. Note that directly training classifiers on the test dataset selects models that tend to never predict failures while achieving accuracies over 96%. Next, we analyze features with the goal to remove those that appear redundant or unnecessary. This step not only accelerates the training but also makes classifiers more robust and explainable.

In Fig. 2(a), we compare features applying the χ^2 test to the training dataset. The highest score is achieved by the average post-FEC BER. This indicates that failures are correlated in time, one failure strongly increasing the probability of observing another shortly after. In contrast, features deriving from powers and temperatures are weakly correlated to the output and thus not interesting. Features deriving from CD and DGD score high but are known to be unreliable when the QoT is below FEC limit [8], and may thus be redundant with the post-FEC-BER. In Fig. 2(b.1), we focus on *Health_1* and *Health_2* through a dedicated scatter plot. The first cluster on the top-left corresponds to cases where failures are dominant, also corresponding to non-zero post-FEC-BER values. In contrast, the second cluster on the top right shows how the *Health_2* feature would allow to correctly predict the failure status despite normal *Health_1* values. The last cluster shows both benefits and limitations of the two features. For that cluster, we observe that some “norm” and “fail” samples are well separated according to the features values. Yet, the same features cannot be used to distinguish the majority of samples from that cluster.

4. Performance Comparison of Machine Learning Models

Within the digital twin (DT), failure prediction is meant to drive the control plane to pro-actively mitigate the impact of a predicted failure, e.g. by rerouting the traffic through a safe path. In the following, we first pre-select default ML models [9] based on their performances on the balanced training dataset. Yet, since false positives (FP) and false negatives (FN) may have strongly different operational costs, we further investigate how the selected models perform in terms of FP and FN counts on the test data, which accounts for the realistic class imbalance.

We first compare in Fig. 3(a) six prediction models on the training dataset. M1 is a medium tree, M2 an efficient logistic regression, M3-5 are ensemble models with different presets: subspace kNN, RUS boosted trees and

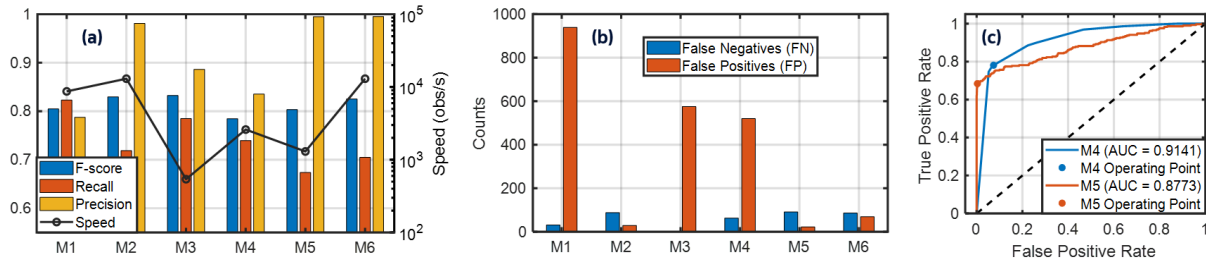


Fig. 3. (a) F-score, recall and precision of selected models on training data (b) False positives and negatives on test data, (c) ROC curves for two models

subspace discriminant, respectively. M6 is a gaussian naive Bayes. Models are detailed in [9]. Models M1-3 leverage all 19 features whereas models M4-6 use only 4 features, marked * in Fig. 3(a). The selected models achieved the best general performances, with F-scores around 0.8. However, we observe that homogeneous F-scores hide large disparities in precision and recall due to the varying amounts of false positives, emphasizing the need for comparing models on the full dataset. The reduced amount of features does not significantly impact performances, including speed. At equal performance, the lower amount of features seems compensated by a higher model complexity.

Then, we plot in Fig. 3(b) FP and FN counts on the full dataset. Models M1, M3 and M4 exhibit large FP counts and low FN counts. In comparison, models M2, M5 and M6 have higher FN counts, but much lower FP counts. The same models have much lower amounts of combined FP and FN. Interestingly, M2 and M5 have comparable performances while M5 only use four features, making it more explainable. Finally, in Fig. 3(c), we plot for comparison the ROC curves for models M4 and M5, obtained by modifying the classification thresholds, and the model operating points corresponding to the performances displayed in Fig. 3(b). With 0.91 against 0.88 AUC, M4 could be considered as a generally better classifier than M5 due to its higher true positive rate. Yet, M5 has a comparable FN count, but a much lower FP count than M4, making it more suitable for an optical network digital twin. With such low FN and FP counts, M5 further achieves a 98.5% accuracy on the full dataset.

5. Conclusion

We studied the prediction of lightpath failures using binary classification to provide digital twins with forecasting ability. We provided a feature analysis and compared the performances of various models on field data from a live production network. Leveraging only 4 features, our preferred model, a ensemble model with subspace discriminant, achieves a 0.8 F-score on balanced data, and 0.89 AUC and 98.5% accuracy on the full data.

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